CGFNet: Frequency-Domain Causal Discovery and Dual-Path Spectral Filtering for Wildfire Prediction

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2025 IEEE International Conference on Big Data (BigData) | 979-8-3315-9447-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/BIGDATA66926.2025.11402223

***Abstract*—Preventing forest wildfires is essential for safeguarding ecosystems, maintaining economic stability, and ensuring public safety. However, existing prediction models often ignore the causal mechanisms underlying variable interactions and fail to capture heterogeneous dynamic behaviors, thereby limiting both interpretability and predictive accuracy. Furthermore, most approaches adopt a separate spatio-temporal architecture, which introduces optimization inconsistencies between modules and neglects the inherent unity of spatio-temporal dynamics, ultimately constraining predictive performance. To overcome these challenges, we propose CGFNet, a unified model that jointly models causal and temporal dependencies for wildfire prediction. Our model maps multi-scale causal relationships identified through Frequency-domain Conditional Mutual Information (FCCMI) into the node space and integrates temporal evolutions within a temporal hypergraph, yielding a unified representation of wildfire causal drivers and sequential dependencies. We introduce a dual-path dynamic filter whose coefficients are adaptively generated from the spectral characteristics of the temporal hypergraph. This design both amplifies causal information and captures heterogeneous dynamic patterns across variables. Subsequently, a multi-band spectral gating (MSG) mechanism adaptively modulates and fuses spectral-band signals on the temporal hypergraph, thereby achieving unified modeling of filtered causal and temporal features. Extensive experiments on the European boreal and mediterranean biomes, using datasets derived from the SeasFire datacube, show that CGFNet substantially outperforms state-of-the-art baselines on wildfire risk prediction metrics. Ablation studies further confirm the essential contributions of both causal discovery and dynamic filtering modules, establishing CGFNet as a robust and interpretable causality-driven solution for wildfire prediction.**

***Index Terms*—Wildfire prediction, causal inference, graph neural networks, spatio-temporal modeling**

I. Introduction

Fire plays a pivotal role in the Earth system, significantly influencing ecosystems worldwide [1]. Wildfires represent a critical natural hazard with far-reaching consequences for societies worldwide, resulting in loss of life, property, infrastructure, and ecosystem services [2]. Hence, it is imperative to improve our understanding and forecasting capabilities of wildfire phenomena within the Earth system. By doing so, we can formulate effective strategies to mitigate the adverse impacts of wildfires and climate change.

In the domain of forest wildfire prediction research, the copious amount of data available lends itself well to sophisticated deep learning (DL) approaches [3] [4], with Convolutional Neural Network (CNN)-based and Recurrent Neural Network (RNN)-based models achieving strong performance. CNN-based approaches typically treat surface data as image snapshots, extracting spatial features of fire risk factors via two-dimensional convolutions, as exemplified by Prapas et al. [5], who formulated burned area prediction as a segmentation task. In contrast, RNN-based architectures such as Long Short-Term Memory (LSTM) [34] and Gated Recurrent Units (GRU) [35], perform autoregressive forecasting on historical burned area sequences, leveraging gating mechanisms to regulate memory decay. Liang et al. [6] developed an LSTM based model for predicting the scale of forest wildfires in Alberta, Canada. However, these two paradigms focus exclusively on either spatial or temporal dimensions, thereby limiting predictive capability.

To overcome the limitations of single-dimensional models that consider only spatial or only temporal information when capturing complex spatio-temporal interactions, researchers

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have proposed integrated spatio-temporal wildfire prediction models. Bhowmik [7] constructed a wildfire database from environmental and historical data, then formulated a specialized U-Convolutional-LSTM neural network, achieving a higher prediction accuracy than conventional CNNs. Malik et al. [8] developed an ensemble model that integrates SVM and XGBoost for processing weather data, alongside CNN and LSTM techniques to handle vegetation and remote sensing image data snapshots, to forecast wildfire risks. These models integrate spatial and temporal information, effectively addressing the limitations of traditional methods while enhancing predictive performance. While having shown the effectiveness of spatio-temporal models, these approaches still face three major shortcomings: (1)most existing models ignore causal relationships between variables. Even studies that attempt causal modeling rely on discrete lag searches with fixed time windows, which cannot capture the multi-scale nature of causal dependencies or the drifting characteristics of continuous time lags. Consequently, they often identify spurious causal pathways, overlook key wildfire drivers, and suffer reduced predictive accuracy. (2) The temporal components of existing models typically employ a uniform structure to process all variables, failing to account for their distinct dynamics (e.g., low-frequency trends in temperature vs. high-frequency fluctuations in precipitation). This homogeneous treatment reduces the ability to capture and respond to critical driving factors. (3) Separating spatial and temporal modeling leads to parameter mismatches and optimization challenges. Moreover, it naturally violates the unified spatio-temporal inter-dependencies in real world, which ultimately hinders the forecasting performance [9].

In this paper, we propose a novel graph neural network architecture, named Causal Graph Filter Net (CGFNet), to address the aforementioned challenges. Our model first identifies multi-scale causal relationships in the frequency domain using the FCCMI method, which extracts the average lag at each frequency scale via phase differences. This process mitigates the effects of lag drift [10] and enables continuous time-delay modeling. Using the causal relationships and the temporal evolution of variables, we construct a temporal hypergraph to uniformly represent inter-variable causal relationships and temporal dynamics. On this temporal hypergraph, a dual-path dynamic filter is applied. The causal path modulates inter-variable interactions, while the temporal path captures the unique spectral profile of each variable. Finally, information is fused on the temporal hypergraph via the MSG mechanism, enabling unified spatio-temporal modeling. The contributions of this paper are as follows:

- We propose a causal identification method that integrates frequency domain analysis and conditional mutual information to identify multi-scale causal relationships among variables and validate them through counterfactual mechanisms, thereby enhancing both the accuracy and interpretability of causal inference.
- We introduces a unified predictive model, CGFNet, for modeling causal relationships and temporal dynamics. Built upon temporal hypergraph structures, the model integrates spectral filtering with multi-channel information aggregation, effectively capturing inter-variable causal dependencies and intra-variable temporal dynamics. This modeling framework enables accurate prediction of complex systems.
- Our model offers superior predictive accuracy compared to existing approaches. Using datasets from the European boreal and Mediterranean biomes, we rigorously benchmark our model against established architectures, validating its efficacy.

II. Related Work

A. Causal Inference Methods for Multivariate Time Series Data

The multivariate synergistic mechanisms underlying forest wildfires reflect genuine causal driving pathways. Accurately identifying the causal structure among variables is of significant importance for revealing underlying driving mechanisms of forest wildfires, enhancing the model's ability to capture synergistic effects and time-delay characteristics, and improving both prediction accuracy and interpretability. Existing research primarily falls into three mainstream paradigms: Granger causality analysis, information-theoretic methods, and state-space modeling. The core of Granger causality methods fundamentally involves assessing whether the historical information of one variable can significantly improve the predictive capability for another variable's future values. Classical methods such as the conditional Granger causality index [11] are widely applied in multivariate time series analysis due to their modeling flexibility and ease of estimation. To address more complex scenarios, researchers have introduced Lasso regularization [12] and kernel methods for nonlinear extensions [13], enhancing the Granger framework's analytical capability for complex systems. In addition, the Granger framework includes the Copula-Granger causal model, which uses the Copula function to describe nonlinear dependencies and is suitable for non-Gaussian distributed systems [14]. Information-theoretic methods exhibit advantages in modeling nonlinear causal relationships. Representative approaches include Transfer Entropy [15] and Partial Symbolic Transfer Entropy (PSTE) [16], which quantify causal strength between variables through entropy and conditional mutual information, making them suitable for highly complex systems such as ecology and meteorology. Informatics methods also cover conditional entropy (CE [17] and partial mutual information (PMI [18], which further expand the dimension of nonlinear causal analysis. CE detects causality by assessing the reduction of conditional entropy, while PMI directly measures information transfer based on conditional mutual information, both of which can effectively handle indirect coupling relationships in multivariate systems. State-space modeling methods capture the drive-response mechanisms and historical dependencies between variables by constructing dynamic evolution models of system state variables (physical or reconstructed manifolds). Classical state-space models [19] are extensively utilized in

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industrial system modeling owing to their clear physical mechanisms and support for parameter estimation. To accommodate chaotic time series analysis, state-space reconstruction methods [20] restore system dynamics through uniform or non-uniform embedding techniques, deriving nonlinear interdependence indices to quantify causal strength. Further developments include Convergent Cross Mapping [21], which detects small-sample nonlinear causal relationships through manifold correlation. However, these traditional methods still face significant challenges when addressing key features such as nonlinear relationships, multi-scale coupling, and the dynamic drift of continuous time lags, struggling to adequately address the demands of complex systems such as forest wildfires. To this end, We proposes the FCCMI method, which organically integrates frequency-domain causal identification with counterfactual validation. This approach effectively decomposes signals, determines causal directions, and eliminates spurious causality, providing a solution for causal modeling that simultaneously ensures physical consistency, frequency sensitivity, and counterfactual verifiability.

B. Forest Wildfire Prediction Based on Graph Neural Networks

Given that real-world data can be naturally modeled as graph data consisting of nodes and edges [22], Graph Neural Networks (GNNs) have shown great promise in geospatial prediction tasks. This capability, which leverages the complex relationships between nodes, has made GNNs instrumental in areas such as forest wildfire prediction. By leveraging their ability to model complex relationships between nodes, GNNs have become instrumental in forest wildfire prediction. GCN-LSTM [23] ransforms global land surface simulation data into graph representations, effectively mitigating data gaps in oceanic regions and enhancing the modeling of long-term dependencies across distant regions. In this framework, the Graph Convolutional Network (GCN) [24] component captures spatial dependencies, while the LSTM handles temporal dependencies, enabling the model to achieve high predictive accuracy across spatial and temporal dimensions. Causal-GNN [25] employs PCMCI [26] algorithm to identify relationships between variables, constructs a corresponding causal graph, and extracts temporal features to represent each node. These node representations are then updated within the causal graph to generate final predictions. Such an approach not only improves predictive performance but also provides deeper insights into the mechanisms underlying wildfire occurrences. However, these models still struggle to capture the heterogeneous dynamics of variables. Moreover, their separate handling of spatial and temporal dimensions restricts cross-domain information interaction, ultimately limiting overall predictive performance.

III. Methodology

In this section, we first give the mathematical definition of the problem we are addressing in this paper. Subsequently, we outline the overall architecture of the proposed model. Finally, we provide a detailed description of the two core stages of our model, multi-scale causal discovery and unified spatio-temporal modeling and prediction, focusing on their workflows and theoretical foundations. This explanation demonstrates the feasibility and effectiveness of the approach for causal identification and multivariate prediction.

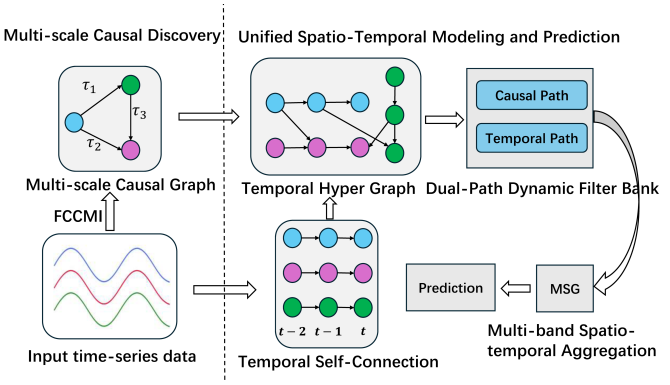


Fig. 1. The framework of CGFNet. The framework comprises two synergistic phases: (1) The Multi-scale Causal Discovery module analyzes input data in the frequency domain, identifies multi-scale causal relationships among variables, and outputs a causal graph with time lags; (2) The Unified Spatio-Temporal Modeling and Prediction module integrates the identified causal structure with variables' inherent temporal evolution to construct a temporal hypergraph. It achieves joint spatio-temporal modeling through a dual-path dynamic filter and multi-band spatio-temporal aggregation, ultimately outputting wildfire risk prediction results.

A. Problem Definition

This study focuses on the modeling and prediction of multivariate time series data within a specific region. The input data takes the form of a 3D tensor $X \in \mathbb{R}^{N\times T\times d}$, where N denotes the number of variables, T represents the time series length, and d indicates the feature dimension at each time step. The core objective is to construct a mapping function $\mathcal{M}$ that predicts future wildfire risk conditions $\widehat{Y}_{(t+1):(t+h)} \in \{0,1\}$ for the next h time steps, based on historical information $X_{(\mathrm{t-s}):\mathrm{t}}$ over the past s time steps from all variables. The output $\widehat{Y}$ is a binary classification label (1 indicating risk present, 0 indicating no risk). This task is formally defined as:

$$X_{(t-s):t} \overset{\mathcal{M}}{\rightarrow} \widehat{Y}_{(t+1):(t+h)} \tag{1}$$

B. Framework of CGFNet

CGFNet is a causality-driven temporal hypergraph neural network framework designed to enhance wildfire prediction accuracy through unified spatio-temporal modeling. The framework is structured around two key phases: multi-scale causal discovery and unified spatio-temporal modeling and prediction.

In the first phase, the proposed model utilizes an FCCMI method to identify causally meaningful relationships between variables in the frequency domain. By applying the FCCMI method, the model uncovers multi-scale lag mechanisms (e.g., delayed temperature effects) and constructs a time-lagged causal graph, thereby identifying the key drivers of wildfire

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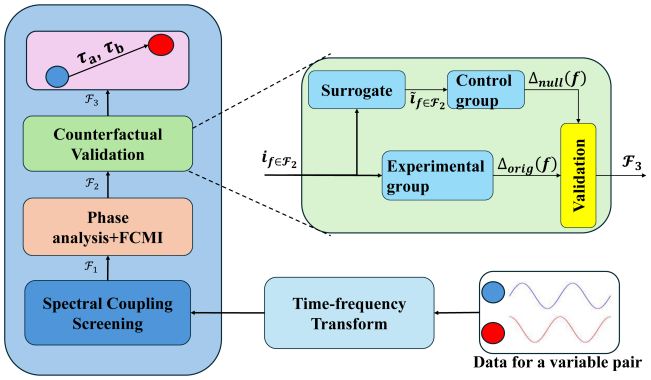


Fig. 2. Overall architecture of FCCMI. (i) Screening for significant coupling frequencies between two variables based on CPSD to obtain frequency set $\mathcal{F}_1$; (ii)Determining the leading relationship through phase analysis and quantifying influence strength via FCMI, followed by identifying frequencies with statistically significant non-zero FCMI to form set $\mathcal{F}_2$; (iii) Employing counterfactual tests to distinguish true from spurious causal relationships, outputting the true causal frequency set $\mathcal{F}_3$, and converting it into time lags. Surrogate denotes the construction of counterfactual signals.

evolution. This provides a robust foundation for interpretable wildfire risk prediction. In the second phase, the discovered causal structure is mapped onto spatio-temporal nodes and combined with the temporal evolution of variables to construct a unified temporal hypergraph, which jointly encodes causal dependencies and temporal dynamics. On this hypergraph, a GraphFilterNet module is introduced, incorporating a dual-path dynamic filter and an MSG mechanism to jointly model inter-variable causal interactions and intra-variable dynamics within a unified spectral-domain graph signal processing framework. This design effectively addresses parameter mismatch, optimization difficulties, and information fragmentation inherent in traditional separate spatial and temporal modeling, thereby enabling high-precision and interpretable wildfire risk prediction.

We train the proposed model using binary cross-entropy loss:

$$L = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)] \tag{2}$$

where $y_i \in \{0,1\}$ denotes the ground-truth label of the i-th sample, $\hat{y}_i \in [0,1]$ represents the predicted probability for the i-th sample.

C. Multi-scale Causal Discovery

The multi-scale causal mechanisms between climate and ocean variables (such as time-lagged drying effects and teleconnection regulation) have been demonstrated to play crucial roles in the development process of forest wildfires [27] [28]. Moreover, intricate causal relationship modeling among variables can support robust and reliable DL [29] [30].To accurately model such causal drivers, the first stage of CGFNet reveals multi-scale causal relationships in the frequency domain via FCCMI, as illustrated in Fig. 2.

1) Frequency-Domain Coupling Screening Based on Cross-Power Spectrum: converting the time-domain data of variable pair (i,j) to the frequency domain, FCCMI first computes their cross-power spectral density (CPSD) to screen for significantly coupled frequencies:

$$S_{ij}(f) = \mathbb{E}_t\left[i_f(t)\cdot(j_f(t))^*\right] \tag{3}$$

where $i_f(t)$ and $(j_f(t))^*$ epresent the spectral component of variable i and the conjugate term of variable j at frequency f , respectively. $S_{ij}(f)$ reflects the synergistic relationship between the two variables at this frequency.

2) Causal Direction and Strength Quantification: In climate and environmental research, phase analysis methods (e.g., wavelet phase analysis) are employed to identify when one variable leads or lags behind another across different frequencies and time periods. This approach helps reveal causal or predictive relationships by demonstrating whether and how changes in one variable precede those in another, as demonstrated by Alekseev [31], who determined that temperature changes lead carbon dioxide concentration variations by 10–15 years. For frequencies exhibiting significant coupling between variable pair (i,j), phase analysis determines the influence direction and time lag:

$$\tau_f = \frac{\arg\left(S_{ij}(f)\right)}{2\pi f} \tag{4}$$

Here, $\arg\left(S_{ij}(f)\right) \in (-\pi,\pi)$ denotes the phase difference of i relative to j, indicating the temporal leading relationship between variables, while τ_f represents the corresponding time difference ($\tau_f > 0$ implies i leads j). The phase difference $\arg\left(S_{ij}(f)\right)$ is essentially a statistic of the time shift of signals at frequency f, which can be directly converted to the physical lag τ_f via (3). Compared to traditional discrete lag search methods, this approach naturally supports continuous lag modeling in the frequency domain. Moreover, it independently estimates the lag for each frequency component, thereby avoiding dynamic drift adaptation biases caused by preset windows.

This study utilizes Frequency-Domain Conditional Mutual Information (FCMI) to directly quantify the strength of causal effects between variables i and j at specific frequencies. By controlling for the confounding variable set Z, this method effectively eliminates indirect associations and precisely captures direct causal dependencies at specific frequencies. Assuming i leads j, FCMI is calculated as:

$$\mathrm{FCMI}(i \to j|Z;f) = H\left(j_f|Z\right) - H(j_f|i_f,Z) \tag{5}$$

where the frequency-domain conditional entropy $\underline{H}(\cdot)$ can be estimated using the K-nearest neighbors (KNN) method based on neighborhood distance. i_f and j_f denote the components of i and j at frequency f, respectively. The construction of the conditioning set Z is guided by CPSD and phase analysis, employing a recursive screening strategy: starting from an empty set, candidate factors that exhibit phase-leading relationships to target variable j are iteratively added to Z. After each addition, if the FCMI value decreases significantly, the factor

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is retained; otherwise, it is considered redundant or indicative of a potential spurious causal path and thus excluded. This process aims to make the FCMI calculation approximate the Markov assumption. That is, when Z contains the Markov blanket of j_f (i.e., its direct causes and confounders), if i_f and j_f are frequency-domain independent (i.e., the FCMI value is statistically close to 0), it can be inferred that no direct causal connection exists between them. This provides theoretical support for FCMI-based causal inference.

3) Counterfactual Validation: Although FCMI can effectively capture statistical dependencies, it may still yield spurious causal inferences in scenarios such as synchronization-driven effects or frequency overlap. To address this, FCCMI introduces a validation mechanism based on counterfactual perturbation: for the original variable i, a surrogate variable $\tilde{i}$ is constructed using the IAAFT (Iterative Amplitude Adjusted Fourier Transform) method [32], which preserves statistical properties while severing causal pathways, satisfying three key properties: preservation of frequency-domain magnitude ($|\mathcal{F}(i)| \approx |\mathcal{F}(\tilde{i})|$), preservation of time-domain distribution ($dist(i) = dist(\tilde{i})$), and disruption of temporal phase structure ($\angle\mathcal{F}(i) \notin \angle\mathcal{F}(\tilde{i})$); by comparing changes in FCMI between the original signal i_f and the surrogate $\tilde{i}_f$ before and after perturbation, true causal relationships are distinguished from spurious correlations.

Specifically, random perturbations are applied to the amplitude and phase of both the original signal i_f and the control signal $\tilde{l}_f$ (amplitude perturbation: uniform distribution within a; phase perturbation: Gaussian noise with variance σ^2), yielding the counterfactually perturbed signal $i_{cf,f}$ and $\tilde{i}_{cf,f}$. To quantify the impact of these perturbations, the absolute changes in FCMI are computed for both the experimental and control groups as follows:

$$\Delta_{\text{orig}}(f) = \left|\text{FCMI}(i \to j|Z; f) - \text{FCMI}\left(i_{cf} \to j|Z; f\right)\right| \quad (6)$$

$$\Delta_{null}(f) = \left|\text{FCMI}(\tilde{i} \to j|Z; f) - \text{FCMI}\left(\tilde{i}_{cf} \to j|Z; f\right)\right| \quad (7)$$

This perturbation is applied to the control group N times to construct a null distribution $\{\Delta_{null}^{(k)}(f)\}_{k=1}^{N}$. If the following condition is satisfied, a causal effect at frequency f is confirmed:

$$\Delta_{orig}(f) > \text{quantile}(\left\{\Delta_{null}^{(k)}(f)\right\}_{k=1}^{N}, 0.95) \quad (8)$$

This threshold corresponds to a significance level of $\alpha = 0.05$, indicating that the observed FCMI variation in the original signal exceeds 95% of the variations expected under the null (control) distribution. In genuine causal relationships, perturbations are expected to disrupt the causal pathway due to strong phase/amplitude dependencies. In contrast, spurious causalities, being primarily driven by statistical coincidences, exhibit similar responses to perturbations in both the original and control groups. Thus, only when $\Delta_{orig}(f)$ significantly exceeds the control group's natural fluctuation range *i.e.*, satisfying (8)) can a true causal path from i to j at frequency f be confirmed.

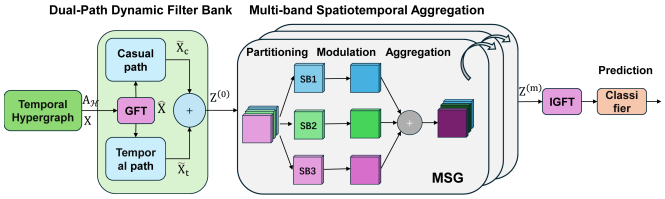


Fig. 3. Overall architecture of GraphFilterNet. GraphFilterNet comprises a dual-path dynamic filter, a multi-band spatio-temporal aggregation module, and a prediction module. The dual-path dynamic filter processes inter-variable causal dependencies and the temporal evolution of individual variables in the spectral domain, respectively. The multi-band spatio-temporal aggregation module fuses spatio-temporal signals across different spctral bands via an adaptive gating mechanism. Finally, predictions are output through an inverse graph Fourier transform and a graph classifier (comprising graph pooling and an MLP). SB denotes the spectral band.

Ultimately, FCCMI integrates multi-frequency identification results to construct a causal graph incorporating frequency hierarchies and continuous time-lag information. Compared to traditional methods based on discrete time lags, FCCMI's phase-difference-based continuous modeling offers enhanced stability and naturally accommodates cross-timestep causal pathways. This provides high-quality input structures for temporal hypergraph construction and underpins GraphFilterNet's unified spatio-temporal modeling.

D. Unified Spatio-Temporal Modeling and Prediction

The second stage of CGFNet employs the GraphFilterNet module to model both causal and temporal information on a temporal hypergraph. This hypergraph serves as the input structure for GraphFilterNet, uniformly representing spatio-temporal dependencies in the form of a directed heterogeneous graph. GraphFilterNet consists of a dual-path dynamic filter, a multi-band spatio-temporal aggregation module, and a prediction module, which collaboratively achieve spatio-temporal dependency modeling and prediction in the spectral domain.

1) Temporal Hypergraph Construction: To address issues such as parameter mismatch caused by separate spatio-temporal modeling, we introduce a temporal hypergraph $\mathcal{G}_{\mathcal{H}}$. The central idea is to unify two dimensions, namely inter-variable causal dependencies and intra-variable temporal dynamics, into a single extended graph structure rather than processing them in isolation as in conventional architectures. The hypergraph consists of $N \times T$ nodes (each representing a variable's state at a specific timestep), with its edge set consisting of two types:

- causal edges capturing asynchronous causal relationships between variables. Specifically, if a causal relationship exists from variable i to variable j with a lag of τ timesteps in the causal graph $\mathcal{G}_c$, a directed edge is

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formed in $\mathcal{G}_{\mathcal{H}}$ from node i_t (state of variable i at timestep t) to node $j_{t+\tau}$, collectively constituting the causal adjacency matrix $A_c \in \mathbb{R}^{(N\times T)\times(N\times T)}$;

- temporal edges that capture individual variables' dynamic evolution over time, derived from temporal self-connections where edges connect consecutive timesteps of the same variable (e.g., from i_t to i_{t+1}), forming the temporal adjacency matrix $A_t \in \mathbb{R}^{(N\times T)\times(N\times T)}$.

The unified graph structure consolidates heterogeneous dependencies, which would otherwise necessitate separate module-specific modeling, into a single graph domain $\mathcal{G}_{\mathcal{H}}$. This design allows the subsequent GraphFilterNet module to directly execute unified graph signal processing on the temporal hypergraph, including graph Fourier transform, filtering, and aggregation. This approach naturally and collaboratively models spatio-temporal coupling while avoiding the communication bottlenecks and inconsistencies inherent in decoupled architectures.

2) Dual-Path Dynamic Filtering: Causal dependencies (inter-variable asynchronous interactions) and temporal evolution (intra-variable dynamics) exhibit distinct spectral characteristics. To effectively capture this heterogeneity while avoiding feature confusion, the model introduces a dual-path dynamic filter: the causal path focuses on inter-variable interactions, while the temporal path concentrates on intra-variable dynamic evolution. Spectral Transformation: based on the singular value decomposition (SVD) of the asymmetric normalized Laplacian matrix $\mathcal{L}_{\mathcal{H}} = I - D_{\text{out}}^{-1} A_{\mathcal{H}} = U\Lambda V^T$ of the temporal hypergraph $\mathcal{G}_{\mathcal{H}}$, the graph Fourier transform for the temporal hypergraph is defined:

$$\widehat{X} = V^T X \tag{9}$$

Here, $X \in \mathbb{R}^{N\times T\times d}$ represents the time-domain feature matrix corresponding to the temporal hypergraph $A_{\mathcal{H}}$, and V is the right singular vector matrix obtained from SVD of the Laplacian matrix. Dynamic Filter Design: in the spectral domain, the proposed model performs filtering operations along both causal and temporal paths based on the causal Laplacian matrix $\mathcal{L}_c$ (derived from A_c) and the temporal Laplacian matrix $\mathcal{L}_t$ (derived from A_t):Causal Path Filtering: The Softplus activation function is employed to ensure non-negative spectral responses and mitigate potential information loss:

$$\tilde{X}_c = Softplus(W_\phi \mathcal{L}_c \hat{X} + b_\phi) \odot \hat{X} \tag{10}$$

Temporal Path Filtering: Utilizes the Sigmoid activation function as a band-pass gating mechanism:

$$\tilde{X}_t = Sigmoid(W_\psi \mathcal{L}_t \hat{X} + b_\psi) \odot \hat{X} \tag{11}$$

Here, W_ϕ, W_ψ, b_ϕ, b_ψ are learnable parameters, $\tilde{X}_c$, $\tilde{X}_t$, represent the filtered outputs of the causal path and temporal path, respectively, where $\odot$ indicates the Hadamard product. The dual-path filter is dynamically derived from the input signal's spectral features $\hat{X}$ (instead of relying on static parameters), allowing context-aware adaptation (e.g., to variable-specific temporal patterns or causal strengths) for fine-grained modeling of multivariate heterogeneous dynamics.

To preliminarily integrate the differentiated features from the causal and temporal paths, the model generates a joint representation through adaptive weighted fusion:

$$Z = W_f \odot \tilde{X}_c + (1 - W_f) \odot \tilde{X}_t \tag{12}$$

$$W_f = \sigma(MLP_1([\tilde{X}_c \oplus \tilde{X}_t])) \tag{13}$$

Where $\oplus$ denotes channel-wise feature concatenation, and $MLP_1(\cdot)$ outputs the fusion weight W_f. This step provides an input representation containing dual-path features for subsequent modules.

3) Multi-band spatio-temporal Aggregation: To accurately model critical spectral components, we introduce an MSG mechanism. This mechanism partitions the spectrum into independent spectral bands on a unified temporal hypergraph and employs learnable gating vectors to modulate and fuse causal and temporal features within each band. This enables unified modeling of spatio-temporal dependencies:

$$Z^{(l+1)} = \sum_{k=0}^{K} \Phi_k^{(l)} \odot (\Lambda_k \cdot Z^{(l)}) \tag{14}$$

The spectral band index $k \in [0, K]$ divides the spectrum into $K+1$ independent bands. The spectral response matrix $\Lambda \in \mathbb{R}^{(N\times T)\times(N\times T)}$, derived from the SVD of the hypergraph Laplacian matrix, represents signal intensities across different bands. Each band k corresponds to a spectral mask Λ_k, constructed by retaining only the diagonal elements of the k-th band in Λ. $(\Lambda_k \cdot Z^{(l)})$ represents the extraction and modulation of signal components within this spectral band. The core of MSG lies in the learnable gating vectors $\Phi_k^{(l)} \in \mathbb{R}^d$, which adaptively modulate the weights of each spectral band via Hadamard product. By aggregating the modulated results across all bands, MSG achieves spectral-aware feature fusion. The entire process follows a 'decomposition-modulation-aggregation' workflow, effectively capturing dynamic structures across multiple spectral bands—including slow-varying trends (e.g., gradual temperature changes), rapid-varying perturbations (e.g., abrupt precipitation shifts), and causal lags.

4) Prediction: After completing m layers of MSG, the node-level spectral-domain feature matrix $Z^{(m)}$ of the temporal hypergraph is utilized for graph-level prediction:

$$\hat{Y} = \sigma(MLP_2(GlobalPool(U \cdot Z^{(m)}))) \tag{15}$$

Here, U denotes the orthogonal matrix of left singular vectors obtained from SVD decomposition of the Laplacian matrix of the temporal hypergraph $A_{\mathcal{H}}$. The operation $U \cdot Z^{(m)}$ corresponds to the Inverse Graph Fourier Transform (IGFT), which maps spectral-domain features back to the node-level time-domain representation.

IV. Experiments

To evaluate the performance of CGFNet, we conducted experiments on two regional datasets from the Mediterranean and Nordic areas, both constructed from the SeasFire Datacube [33]. The SeasFire datacube, which comprises 59 variables

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encompassing climate, vegetation, oceanic indices, and human factors, has an 8-day temporal resolution and a spatial resolution of 0.25 deg, and spans from 2001 to 2021. It is designed to support global subseasonal-to-seasonal wildfire modeling and prediction, covering key elements such as atmospheric parameters, land cover, and biomes. We select three local weather variables Temperature at 2 meters - Mean (T2m), Total precipitation (TP), Vapor Pressure Deficit (VPD), and three Oceanic Climate Indices (OCIs) known to influence the test area, including North Atlantic Oscillation (NAO), Arctic Oscillation Index (AO), and Nino 3.4 Anomaly (Nino34_anom) as predictors for wildfires. These variables are widely recognized in wildfire research as the dominant factors affecting fuel moisture, atmospheric drying, and climate variability, capturing both local fire-weather conditions and large-scale climate influences. The target variable is a binary map of the burned areas. The causal graph contains seven nodes with the target variable included. In the temporal hypergraph, edges and nodes associated with the target variable are masked to prevent label leakage. The model uses 2003 to 2015 data for training, 2016 to 2017 for validation, and 2018 to 2019 for testing.

A. Baselines

On the aforementioned datasets, we compare CGFNet against the following models:

- LSTM [34]: A classic Long Short-Term Memory network that effectively captures long-term dependencies in time series through gating mechanisms (input gate, forget gate, output gate), widely used for sequential modeling and prediction.
- GRU [35]: Gated Recurrent Unit, a simplified variant of LSTM that merges partial gating structures (update gate and reset gate), typically achieving comparable performance with fewer parameters and higher computational efficiency.
- Causal-GNN [25]: A causal graph neural network architecture that explicitly models inter-variable causal relationships while integrating temporal features for prediction.
- T-GCN [36]: A Temporal Graph Convolutional Network model that integrates Graph Convolutional Networks (GCN) with Gated Recurrent Units (GRU) for spatio-temporal modeling.

B. Experimental Setup

The training and validation sets are downsampled at a 5:1 negative-to-positive ratio, while the test set retains the original distribution. For MSG (number of spectral bands = 6), we configure 5 layers with an input time window of 90 steps. The amplitude perturbation range a is set to 0.1, perturbation phase variance $\sigma = 0.05$, and counterfactual perturbation count $\mathbf{N} = 100$. We employ the Adam optimizer to train our model for up to 100 epochs, with an initial learning rate of 0.001. Missing values are directly excluded as mentioned above. The evaluation metrics we choose the Area Under the Precision-Recall Curve (AUPRC) and the Area Under the Receiver Operating Characteristic Curve (AUROC).

TABLE I
AUPRC AND AUROC PERFORMANCE OF THE DIFFERENT MODELS FOR H = 8 DAYS IN MEDITERRANEAN.

Model	AUPRC(%)↑	AUROC(%)↑
Random Classifier	1.1	50.0
LSTM	32.9	91.9
GRU	34.6	91.6
T-GCN	34.3	92.2
Causal-GNN	34.6	92.4
CGFNet(ours)	**34.8**	**92.9**

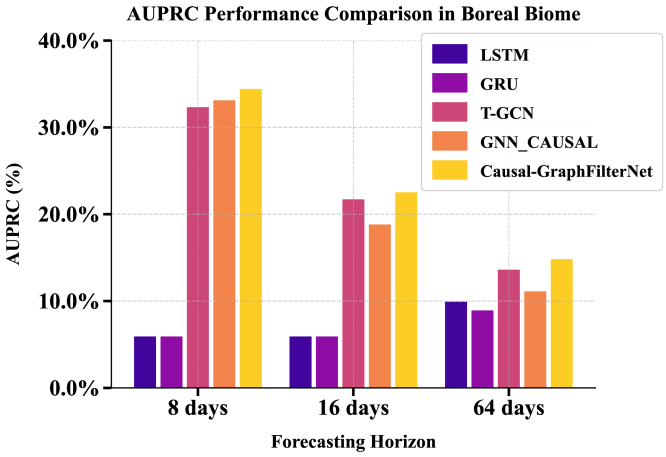


Fig. 4. AUPRC performance of the different models at forecasting horizon of 8, 16, 64 days in Boreal.

C. Experimental Results

1) Model Performance: Table I compares the performance of CGFNet with that of baseline models for 8-day-ahead wildfire risk prediction (in terms of AUPRC and AUROC) in the Euro-Mediterranean region. As shown in Table 1, CGFNet outperforms all baseline models on both metrics. LSTM and GRU models, in contrast, primarily focus on temporal modeling and fail to explicitly capture complex inter-variable interactions, which constitutes a key limitation in their performance. Although T-GCN and Causal-GNN incorporate spatio-temporal modeling, their performance still falls short of that of CGFNet. This may be attributed to their separate spatial and temporal modeling. This design may face parameter optimization compatibility issues during joint training and exhibits limitations in comprehensively capturing complex cross-spatio-temporal coupling dependencies. In contrast, the superior performance of CGFNet stems from its unified architecture enabling refined modeling and comprehensive learning of causal dependencies and temporal dynamic evolution features.

As illustrated in Fig. 4, CGFNet also surpasses baseline models in the Northern region. However, despite its strong short-term performance, the model's accuracy degrades noticeably as the prediction horizon extends. This limited long-term prediction capability may relate to its reliance on a global static causal graph. In climate systems, causal mechanisms

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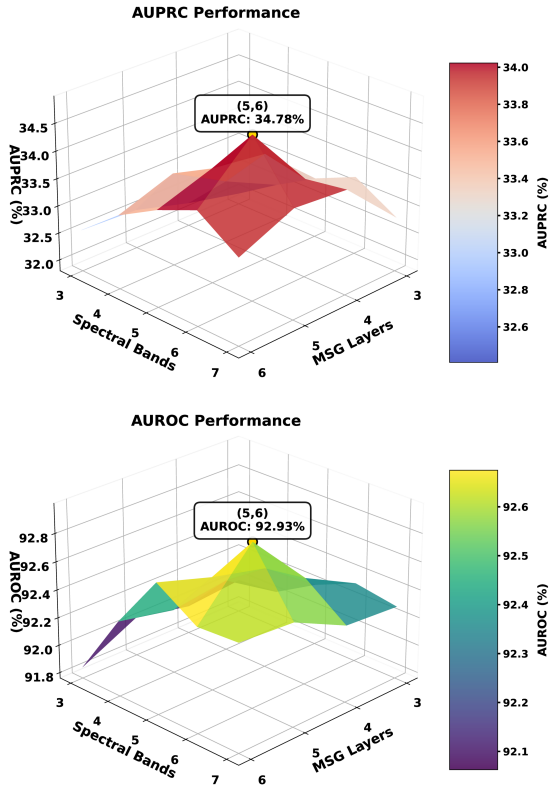


Fig. 5. Radar chart showing the effect of MSG layers and spectral bands on the model's AUPRC performance.

often shift across temporal scales or environmental states. For instance, the impact of precipitation on wildfires manifests as the direct effect of current soil moisture in the short term, while in the long term, it emerges through the indirect mechanism of drought accumulation. Under such conditions, a static causal graph fails to adapt to dynamic shifts in causal pathways, potentially overlooking key mechanisms or introducing misjudgments. This highlights a current limitation of the model and suggests a direction for future work (e.g., exploring dynamic causal graphs).

Through systematic parameter sensitivity experiments, we found that the model achieves optimal performance on the validation set when the number of MSG layers (L) is set to 5 and the number of spectral bands (K) is set to 6. This result reveals three key mechanisms: First, the 5-layer MSG structure optimally models cross-band interactions by balancing the trade-off between shallow feature extraction and deep feature fusion. Second, 6 spectral bands cover the primary spectral range of meteorological signals (consistent with the Nyquist sampling theorem), avoiding redundant computation. Finally, the dual-path filter exhibits significant synergistic effects under this parameter configuration.

2) Causal Graph: To evaluate the effectiveness of the FCCMI method and the physical interpretability of its outputs, we present the identified causal topological structures in the

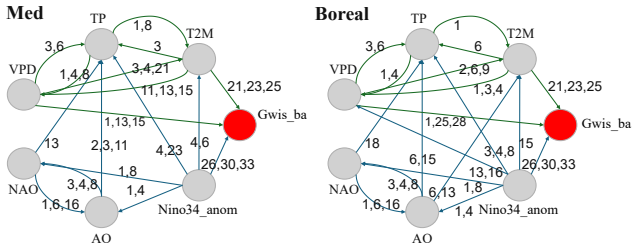


Fig. 6. This figure illustrates the causal topological structure among variables in the Euro-Mediterranean and Nordic regions. The value labeled on each edge represents the time lag of the causal effect (unit: cycle; 1 cycle = 8 days). Blue edges represent edges originating from ocean variables, while green edges represent edges originating from weather variables.

Euro-Mediterranean and Northern regions (Fig. 6). This figure clearly illustrates significant causal dependencies between variables and their corresponding multi-scale temporal lags.

Potential causal relationships between variables are visualized in the causal diagram (as shown in Fig. 6). Vapor pressure deficit (VPD) exhibits significant bidirectional effects on precipitation (tp) at a 0-month lag [37] [38], while also demonstrating bidirectional coupling with temperature changes—manifesting not only significant contemporaneous effects (0-month lag) but also sustained significant impacts at a 21-period lag (approximately 5.5 months). The analysis further reveals strong bidirectional causal relationships between AO and NAO within 1-4 cycles (0-month), a finding consistent with the research outcomes of Ambaum et al. [39]. Notably, Nino34_anom exhibits differential impacts across ecological regions: directly influencing precipitation patterns in the Mediterranean region while primarily affecting temperature variations in northern forests, aligning with observations from Benassi et al. [40]. Consistent with expectations, AO emerges as a key driving factor: significantly influencing precipitation dynamics in Mediterranean forests while serving as the primary regulator of temperature variations in northern regions. Meanwhile, NAO, a critical factor affecting Mediterranean precipitation, exerts impacts with a lag of 13 periods (approximately 3.5 months) [41]. These identified causal relationships align with theoretical understanding and empirical consensus in climate system dynamics and wildfire ecology, confirming the effectiveness of the FCCMI method.

3) Ablation Study: To evaluate the critical contributions of the proposed causal graph discovery module (FCCMI) and dynamic frequency-domain filter bank to model performance, we designed the following ablation studies as controlled experiments:

- CORR Graph Filter Net: Replaces the FCCMI method with a simplified version that constructs the temporal hypergraph adjacency matrix using the Pearson Correlation Coefficient (CC). This adjacency matrix is computed from node features.
- Causal Graph Net: Removes the dynamic frequency-domain filter bank module (i.e., (10)-(11)) from CGFNet while retaining all other components.

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TABLE II
COMPARATIVE EXPERIMENT RESULTS OF CGFNET AGAINST MODELS USING TEMPORAL HYPERGRAPH ADJACENCY MATRICES CONSTRUCTED VIA CORRELATION COEFFICIENTS AND MODELS WITH DYNAMIC FILTERS REMOVED.

Model	AUPRC(%)↑	AUROC(%)↑
CORR Graph Filter Net	31.5	90.6
Causal Graph Net	31.2	89.3
CGFNet(ours)	**34.8**	**92.9**

The ablation study results (Table II) demonstrate that CGFNet significantly outperforms both the model constructing adjacency matrices via correlation coefficients (CORR Graph Filter Net) and the model with the dynamic filter bank removed. The performance limitation of CORR Graph Filter Net primarily stems from its reliance on statistical correlations, which fail to distinguish true causality from spurious correlations. This leads to the introduction of numerous spurious connections in the constructed temporal hypergraph, severely interfering with the model's learning of critical wildfire driving mechanisms. Conversely, the model with the dynamic filter bank removed (Causal Graph Net) faces dual challenges: On one hand, it is inherently difficult to disentangle causal dependencies from the temporal evolution of variables, as they represent fundamentally different types of relationships; on the other hand, it loses the capability to perform fine-grained modeling of distinct spectral dynamics across different variables (e.g., low-frequency trends in temperature versus high-frequency fluctuations in precipitation), significantly weakening the capture of temporal features. These results robustly validate: (1) the importance of constructing causal relationships between variables for wildfire prediction tasks; (2) the critical role of the proposed dual-path dynamic frequency-domain filter in effectively distinguishing and enhancing causal and temporal modeling.

V. CONCLUSION

To address the fundamental challenges of spatio-temporal information fragmentation, dynamic feature confusion, and spurious causal interference in traditional forest wildfire prediction, we propose the CGFNet model. For the first time, the proposed model integrates multi-scale causal relationships, dynamic filtering, and cross-band modulation within a unified temporal hypergraph architecture. This design enables adaptive and fine-grained modeling of key driving mechanisms and substantially enhances cross-domain information interaction. Experiments on real-world datasets demonstrate that CGFNet outperforms benchmark models on key metrics, validating the effectiveness of frequency-domain causal modeling in complex ecosystems. Future work will proceed along two directions. First, the framework will be extended to early-warning tasks for climate disasters such as floods and droughts. Second, adaptive dynamic causal mechanisms and hypergraph structure compression techniques will be developed to improve the model's responsiveness to abrupt environmental changes and to enhance its real-time inference efficiency on a global scale.

ACKNOWLEDGMENTS

This work is supported by the Humanity and Social Science Foundation of Ministry of Education (24YJAZH022), the National Natural Science Foundation of China (62576198, 62476157, 62432006, 62576371, 62576201), the Key Research and Development Program of Shanxi Province (202302010101007), and the Fundamental Research Program of Shanxi Province (202303021221055).

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